

IE1005 – FROM COMPUTATIONAL THINKING TO PROGRAMMING

AY26/27 SEMESTER 1 TAKE-HOME ASSIGNMENT 1

Instructions

- Save your source code as `HW1_YourName.c` and submit only `HW1_YourName.c` in NTULearn.
- You are required to submit a video of yourself explaining the assignment solution. Refer to the instructions on video recording available in NTULearn.
- You may resubmit as many times as you need until the deadline. Only the last files submitted before the deadline will be graded.
- Ensure you submit the correct files. You will be graded based on the files that you submit.

Question

Write a C program that takes the **integer marks** of three students as input, calculates the letter grade for each student, and displays the results in a specified table format. Grades are assigned based on the following criteria:

80 to 100 marks: A

50 to 79 marks: B

0 to 49 marks: C

Requirements:

1. **Strictly NO Decision Making:** You are absolutely prohibited from using any logical decision-making structures, including if-else, switch-case, or the ternary operator (?:) Processing.
2. **Restricted Elements:** The use of arrays, loops, or pointers is strictly forbidden.
3. **Table Formatting:** You must use format specifiers to output the table with spacing and alignment similar with the sample execution.
4. **Algorithm Explanation (Comments):** You must provide detailed code comments explaining the mathematical algorithm used to convert the marks into letter grades (A, B, C) using only mathematical expressions.
5. **Video Explanation:** Your submitted video recording must include a clear verbal explanation of how your decision-free grade conversion algorithm works.

Hints:

1. In C, integer division (`int/int`) truncates the decimal part and returns an integer (e.g., `78 / 50` returns 1).
2. The character data type (`char`) internally holds an ASCII integer value, meaning you can perform arithmetic operations on characters (e.g., `'C' - 1` results in `'B'`).

Note

- You may assume that the user always provides valid input.
- Ensure your program is efficient and easy to read.
- Style your code accordingly.

Sample Output (Bold indicates user's input as an example)

Enter the integer marks of 3 students (e.g. 35 89 53): **99 34 67**

```
=====
Student      Mark      Grade
-----
      1         99        A
      2         34        C
      3         67        B
=====
```

Marking schedule

- Proper declaration of variables 5 marks
- Prompt the user and take the input 15 marks
- Display the results in a table in a suitable format 15 marks
- Explanation of algorithm used to convert the marks into letter grades in comment 10 marks
- Keep good coding style (indentation, headers, variable names, etc.) 5 marks
- Video recording of code explanation and demo (in 3 min) 50 marks
- Total 100 marks
- Using any logical decision-making structures, including if-else, switch-case -30 marks

END